BlindSpot: Qualitative Sensing-Gap Reasoning with a Deterministic Core, a Framing Model, and Hardened Verification

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(author line placeholder for first draft)

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Abstract

Many operational failures hide inside sensing gaps: the days a peat fire burns under cloud, the hours a river valley is unwatched between satellite passes. BlindSpot is a system that reasons about what such gaps may conceal. Most operational systems interpolate what was seen; BlindSpot reasons about what was not seen — the days lost to cloud and the ground between clear views — in space and time together, which is the capability few current systems offer. Its core is deterministic and qualitative: detections become dated grid cells or mapped assets, a hand-written ledger of facts is scored by frozen human weights into a ranked set of mechanism stories, and every conclusion carries the spatial and temporal relations it came from. A language model is allowed exactly one job — to frame studies and draft rival stories from anonymized evidence — and a default-deny verification gate decides what reaches the brief. Nothing the model writes can move a frozen number. Two field studies carry the claim: a 2026 Nepal flood episode, where the frozen brief named impoundment-then-burst at 0.94 and a quarantined ground-truth window confirmed it, and a 2026 Kalimantan peat-fire episode, where the brief named suppression-then-restart at 0.84 and both a hidden time window and a masked half-map confirmed it. A tiered memory layer records past episodes and stays provably silent until five labelled cases share one pattern; a self-evolving agent may propose rule changes, test them on held-out history, and then wait for a human. A morning agent runs the daily round on the operator's own machine — pull, frozen study, memory lookup, preregistered triggers, and two human lines assembled from ledger values, with no model writing a word. Twenty-four hardened gates enforce all of this, and they have caught real defects — from a shifted composition table to a vacuous-truth hole that let empty results pass. This draft reports the misses honestly: one-sided history, an empty flood memory, a pending flood spatial check, and the limits of what gates can see. It closes with the improvements that would make future demonstrations ones other systems cannot easily copy.

1 Introduction

An operator watching a hazard — a flood spreading down a valley, a peat fire smouldering under haze — rarely sees it continuously. Satellites pass overhead on schedules of their own, clouds cover the ground for days, and between one clear observation and the next there is a gap. The most dangerous possibility is often the one that fits entirely inside that gap: a dam that fails between passes, a suppressed fire that restarts under smoke. BlindSpot exists to reason about exactly that possibility, and to do so in a way an operator can audit down to the last arithmetic step.

Three ideas hold the system together. The first is that the reasoning should be *qualitative*. Rather than simulating water or fire physics, BlindSpot works with relations: this region *overlaps*

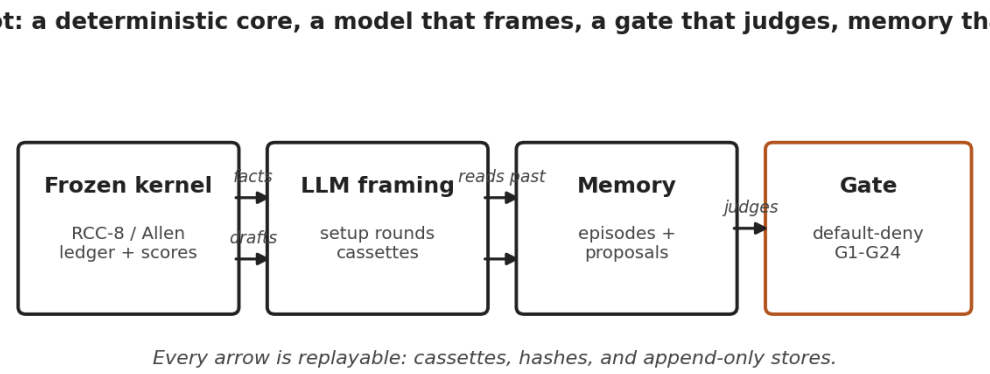


Figure 1: Four parts, one rule. The kernel never learns, the model never scores, memory stays silent below its bar, and the gate is the only judge. Everything the arrows carry is replayable from hashes and append-only stores.

that one, this burn period *meets* that quiet gap, this asset lies in the *near* band of the flood outline. Relations are coarser than numbers but they survive noisy sensors, and — crucially — they compose: from "the lake overlaps the channel" and "the channel meets the village" the system can derive what must, may, or cannot hold downstream, with each step checkable by hand.

The second idea is a strict division of labour between the deterministic core and the language model, best pictured as a measuring tape and a second doctor. The kernel is the tape: it never stretches, never learns, so any brief can be recomputed bit-for-bit years later. That rigidity is deliberate, and it has a price (Section 11 records where the tape had to be amended). The model is the second opinion in the chart: it drafts rival stories from anonymized evidence, the brief prints its disagreement alongside the frozen scores, and the official numbers never move toward it — the lab results decide the treatment, but the second opinion is on the page, not hidden. When model and ledger disagree, as they do in both studies here, that split is the design working, not failing.

The third idea is that verification should be mechanical, not aspirational, because fluent machine-written code fails silently: it produces results that look structurally perfect while being wrong. Twenty-four hardened gates run over every change, in two different moments. Some check the code as it is written (does every relation pair invert exactly? is builder code free of scenario answers?). Others check the system as it runs on real data (hide all raw inputs: does the brief come back identical? replay frozen studies: are the bytes unchanged?). Section 10 explains both moments, names what each caught, and admits what no gate can see.

The running picture is Figure 1: a frozen kernel, a framing model, a memory that mostly stays silent, and a gate that judges everything.

2 The core concept: what a blind spot is

A *blind spot* in this project has a precise meaning, and it comes in two halves that share one cause. The cause is cloud cover, orbit schedules, and the ordinary intermittence of satellite watching: the ground goes unseen, and when it reappears, something may have happened in between.

The *temporal* half: take two observations of the same place at two times, t_0 and t_1. Between them lies a gap of hours or days with no usable observation. A temporal blind spot is a physically possible event sequence that (a) is consistent with everything seen at t_0, (b) is consistent with

everything seen at t_1, and (c) would change what an operator should do — a dam that fills and bursts between passes, a suppressed fire that restarts under smoke.

The *spatial* half: the same clouds hide *places*, not only days. Between a clear upstream view and a clear downstream view lies an unobserved stretch of valley. A spatial blind spot is the ground a hidden process could have reached unseen. The system's spatial answer is two regions: the *must-be* region (ground the process must have touched, given the observations) and the *may-be* region (ground it could have reached at the maximum spread rate over the gap hours). Every asset of interest is then sorted against those regions: inside must-be, inside may-be, or outside both. That sorting is the operator's product — it says which bridges may be out, which power stations are exposed, which helipads remain usable.

The output vocabulary is deliberately small: the two regions, the asset sorting, and a ranked list of mechanism stories with probabilities. In the Nepal flood study that meant a 172.8 km downstream may-be band, 195 exposed assets (171 bridges, 14 hydropower facilities, 10 helipads), and six ranked stories. In the Kalimantan fire study it meant a dated grid of burning cells, continuity judgments between consecutive days, and four ranked stories.

Whether the demonstrations fulfil this concept is a question we answer study by study in Section 5, and the honest summary is: the fire study fulfils it twice over (two independent holdouts), the flood study fulfils it once with the second check openly pending, and the live fire page fulfils only the weaker claim of "reading today," not prediction. A system that cannot say which of its checks are missing is not an operator's system; saying so is part of the design.

3 The algebra: qualitative space and time

The kernel reasons the way a person giving directions does: not in coordinates but in relationships — "the lake overlaps the channel, the channel runs into the village, so the village is exposed." This section explains the four families of relationships it uses, in plain words first and exact conventions second. The conventions matter because half the project's defects turned out to live in them (Section 10): whether "ends Monday, starts Monday" means touching or not, whether every comparison is checked in both directions, and which layer answers "which way" when the operator asks where to look.

3.1 The four relation families

RCC-8 topology. Given two regions, exactly one of eight topological relations holds: disconnected (DC), externally connected (EC), partially overlapping (PO), equal (EQ), tangential proper part (TPP) and its inverse (TPPi), non-tangential proper part (NTPP) and its inverse (NTPPi) [2]. BlindSpot classifies region pairs with exact geometric predicates; any coordinate tolerance is applied once at ingestion from the frozen config, never inside the classifier. Every classification is produced in both directions, and the inverse must be the exact converse — a property the test suite asserts on every pair, because an early version of the composition table had its rows shifted by a sorting mistake and every downstream inference looked structurally fine while being wrong.

Allen intervals. Given two time intervals, exactly one of thirteen relations holds: before, meets, overlaps, starts, during, finishes, equal, and their six inverses [1]. The kernel derives them from endpoints only, on half-open day-resolution intervals, with one load-bearing rule: an end equal to the other's start is MEETS, never BEFORE. Dates are days, not instants, and that single convention decides whether a burn period that ends Monday and a gap that starts Monday touch or not.

Rectangle algebra and tiles. For projected extent reasoning the kernel uses the rectangle algebra [3]: the Allen relation on the x-axis paired with the Allen relation on the y-axis. Rectangles say how extents overlap; they give only coarse relative position. Fine direction — north, northeast, bearing, "which way to look" — comes from the tiles layer beside them: each feature records which tiles of the reference's bounding frame it touches (north, south, east, west, the four corners, and overlap). Tiles are what the map draws; rectangles are what the reasoning checks; together they cover orientation.

Bands. Distances collapse into five preregistered bands: CONTACT (gap of zero), IMMEDIATE_ENVELOPE, NEAR, MEDIUM, and FAR, with the kilometre limits frozen in configuration before any computation. A null in the config means "not decided" and raises rather than defaulting — a rule that exists because a silent default is how scenario answers sneak into builder code.

3.2 Snapshot, propagation, and the gap auditor

The kernel's central verb is *snapshot*: snapshot(A, B) is the relation of A with respect to B, always computed forward (feature-against-reference) and inverse (reference-against-feature), with the inverse required to be the exact converse in every layer. Composition tables let relations chain: knowing A-to-B and B-to-C constrains A-to-C. The neighbourhood graph over the eight RCC-8 relations — which relations can transition directly into which — is not hand-written but derived, with witness counts stored beside it, after the generator proved that a coarse grid made strict containment unrepresentable and forced an explicit two-scale design with teleport jumps (DC to PO, NTPP to DC) verified absent.

The gap auditor turns these relations into the blind-spot vocabulary. From the last-seen geometry, the gap hours, and a maximum spread rate it builds the must-be and may-be bands; it classifies assets against them; and it runs two detectors. Detector A (flood) tests rival upstream/downstream hypotheses constructed from t_0 evidence alone — enforced by the function signature, which simply has no t_1 parameter, so reading the future is a type error rather than a review catch. Detector B (fire) grids detections into dated cells, audits day-to-day continuity per cell, and flags components that vanish and reappear: the formal shape of "suppression, then restart."

3.3 From facts to a ranked brief

Each study's ledger is a small catalog of facts (six fire facts FF1–FF6 in the Kalimantan study: run length, gap structure, clumping, and the like), each either active or inactive for the case at hand. Human-authored rules assign weights; each mechanism story accumulates the weights of the facts that support it; the totals are normalized to probabilities rounded to four places, with a largest-residual correction so they sum to exactly one (a 0.0001 surplus failed the plan's own contract once, and is now fixed in the artifact rather than the tolerance). The frozen fire ranking is shown in Figure 3; the flood ranking appears in Table 1.

Where this sits. The relation families are textbook qualitative spatial and temporal reasoning in the lineage surveyed by Cohn and Hazarika [5]: regions and connection after Randell et al. [2], intervals after Allen [1]. What that literature rarely does is refuse to interpolate: snapshot ontologies in the SNAP/SPAN spirit [6] describe what is seen at successive instants, while the gap auditor reasons about what happened between the snapshots, and the holdouts score whether it was right. Against neural-symbolic systems in the Garcez tradition [7], the division of labour here is deliberately lopsided: the model proposes vocabulary, never numbers, and a deterministic gate

holds the only vote — learning to speak, rather than learning to calculate, which is why the memory layer earns trust by staying silent.

4 What the language model does, and what it is forbidden from doing

The studies in Section 5 are real 2026 emergencies: a flood cascade in Nepal's Langtang valley and a peat-fire episode in Kalimantan. In each case the question put to the language model was the same one a consultant would get on day one: here is the landscape (bridges, dams, burning grid cells) and here are the pre-cutoff facts with all names, dates, and outcomes stripped out — what stories could explain this, and how would you weight them? The model's role is framing without conviction: it may propose the stories, but the scores come from elsewhere. Given a new study, a setup round shows the model the domain ontology and the anonymized pre-cutoff ledger — never names, never dates, never ground truth — and asks it to draft mechanism stories with weights. The round is recorded end to end: prompts, drafts, rejections, and model-usage rows land in append-only stores, and every model call replays offline from committed cassettes (986 model cassettes under a 992-file manifest), so the full gate suite exercises the neuro-symbolic path deterministically with no key and no network.

Three properties keep this honest. First, *knowledge cleanliness is tested, not assumed*: a probe asks the model about the event from anonymized evidence, and the shipped flood probe returned "no knowledge" with a self-stated cutoff predating the flood. Second, *the gate is default-deny*: every number in every model-written sentence must already exist in the ledger's annotated numbers, with entity identifiers masked before the scan (an early version of the rule criminalised the digits inside `bridges:7`). Anything unverified is stripped or the brief is blocked; a skipped check counts as failure. Third, *disagreement is the expected outcome and is published*: the flood model ranked safe drainage first while the ledger put impoundment-then-burst at 0.94; the fire model topped `smoldering_restart` against the frozen `suppression_then_restart`, yet independently weighted the same two lead facts highest (unbroken run 0.95, hotspot clumping 0.9) as the human rules. That split is the design working: the model brings vocabulary, the frozen weights bring conviction, and the ledger arithmetic decides. Table 2 records both disagreements.

The setup history also contains the project's cheapest lesson in format discipline. In an early round the model wrote short ledger identifiers while the checker demanded full ones, producing thirty-odd citation errors; a single instruction line fixed it permanently. A recurring failure in traces, turned into one targeted harness change, with the evidence kept — that is the hill-climbing loop of Section 7 performed once by hand before it was ever automated.

5 The two demonstrations, and whether they fulfil the concept

5.1 Kalimantan peat fire: fulfilled twice

Peat is the fuel that lies. It burns underground, it is declared suppressed, and then it restarts — and the restart is the dangerous moment, because crews have stood down. In mid-September 2026 the satellites watched a Kalimantan peat episode through exactly this shape: 6,142 FIRMS detections over five days (Figure 2, left) across 30 grid cells, 28 of them hot, with a clean multi-day gap in the middle followed by an unbroken burn run. The frozen brief read that shape the way an experienced fire officer would: suppression first, then peat restart — ranked first at 0.8425 against 0.1140 for continued spread (Figure 3, left), before seeing anything that came after.

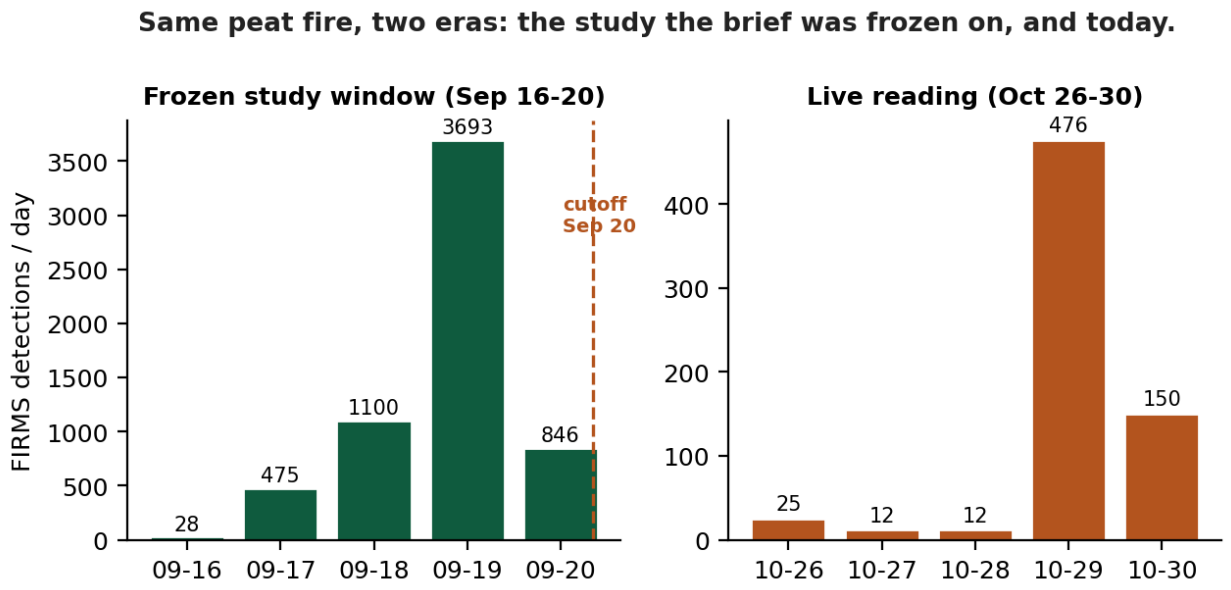


Figure 2: The same fire in two eras. Left: the September window the brief was frozen on (6,142 detections over five days). Right: today's live reading scored by the same frozen rules (675 detections, spike of 476 on October 29, 150 on October 30). The cutoff line marks what the brief was allowed to know.

It was then tested the way a claim like that must be tested: twice, and unseen both times. For the temporal check, every detection after September 20 was locked away before the brief was written; the hidden late-September days confirmed the restart. For the spatial check, the same frozen rules were recomputed over the western half of the map only (3,071 of 6,142 detections kept): the ranking came back identical, probabilities equal to four decimals — identical because scoring depends on which facts are active, not their magnitudes. The activation thresholds behind that sentence were not fitted to the September window: count above zero, five distinct burn days, clumping at three or more, the noise floor of three, all sit in the frozen study rules, which mirror preregistered decisions D-1 through D-6 tagged `prereg` before any pipeline ran. Stating the other edge of that sword too: binary actives buy pervasiveness at the price of magnitude blindness — a thin clump and a vast one both read simply "active," so the holdout proves the shape is everywhere, never how much of it there is. Every gate still passes on half the map, which is itself the finding: the fire signal is spatially pervasive, not one hotspot's doing. Figure 4 summarises all four holdout cells.

A live page extends the study to the ongoing fire without extending its claims. Five October days (26–30) pulled from FIRMS under operator keys score identically under the frozen rules, and the live signature lands in the Iberia-led shape at four labelled cases of five — so the counts tier stays silent for today's reading, while the page names the four matching windows it can actually show (four September windows over Iberia, with dates and what followed). The page states its boundary on its face: this is a reading of today, not a forecast of tomorrow, because no forecasting model exists here and none was invented. Its November outlook is conditional arithmetic, not prediction: *if* the dry regime returns, the validated pattern restarts rather than dies out; *when* (Section 10.1) rain arrived instead, the outlook named rain as its own falsifier in advance.

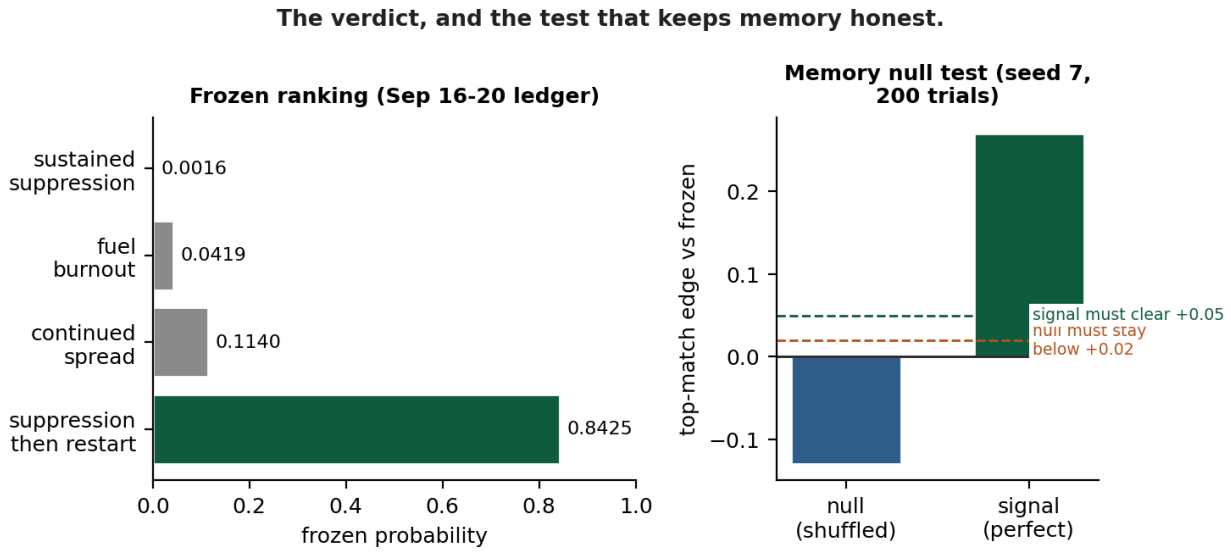


Figure 3: The verdict, and the test that keeps memory honest. Left: the frozen fire ranking. Right: shuffled labels must not beat the frozen top (observed −0.130, bar +0.02), while perfect-signal labels must gain clearly (observed +0.270, bar +0.05). Both margins held with room to spare.

5.2 Nepal flood cascade: fulfilled once, second check pending

A valley floods: bad, but understood — water comes down, bridges wash out, crews respond. What the September 2026 Langtang-valley analysis found was a second disaster hiding inside the first. Upstream, a secondary dam had been impounding water behind a blockage; then it burst, and the burst multiplied the destruction downstream far beyond what the rain alone explains. That is the insight of this study, and it is why the stakes read the way they do: 171 bridges (no bridges, no rescue), 14 hydropower facilities (no power stations, no power), 10 helipads (the last resort when roads are gone) — 195 assets sorted against a 172.8 km downstream may-be band, each one told whether the water must have reached it or merely could have.

The frozen brief named that story — impoundment-then-burst — at 0.9426 (Table 1), before the ground truth was opened. The temporal check then quarantined everything after the early-September cutoff and read the hidden delta only after the freeze: 754 newly destroyed features in the unseen window, in the channel-adjacent pattern the burst story predicts — a MATCH, reported as-is with no re-tuning afterward. The spatial check does not exist yet. Flood damage exports carry almost no dates (two real damage days against 8,368 dateless features), so no masked re-run has been constructed; a repository-wide search verified the absence three ways before it was declared missing rather than assumed present. That asymmetry is a finding about the data, not the method: the flood story is confirmed in time but untested in space, and Section 12 lists exactly what would close the gap.

6 Procedures: step by step, all three demonstrations

This section is the lab notebook: every step of each demonstration in order, with the command, the inputs and outputs, the algebra operation used, and every language-model call. A reader who follows it reproduces every number in Section 5.

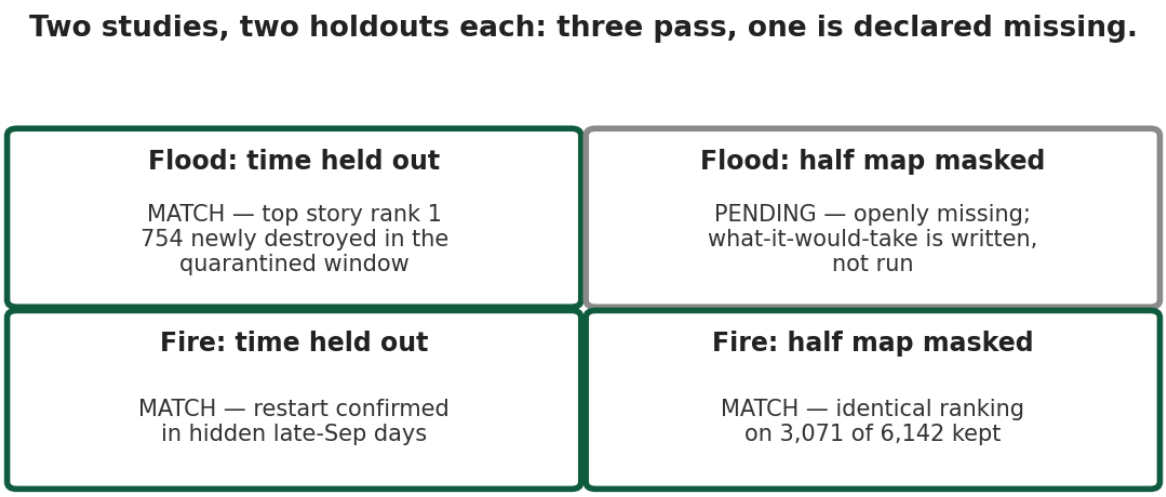


Figure 4: Three of four holdout cells pass; the fourth is declared missing, with the masked re-run specified as what-it-would-take rather than invented.

Step 0 (shared). Freeze the world before touching it. The configuration carries a `prereg` git tag (gate G9 blocks pipeline work without it); every input file is hash-pinned in a manifest (992 files, tool-registered, never hand-edited); any undecided config value is null and raises rather than defaulting. Ground truth is read only after the freeze, reported as-is, never tuned against.

6.1 Nepal flood: curate to ground-truth check

Step 1 — curate. Raw exports become the study inputs (`extent.json`, `target_reach.geojson`, `waterways.geojson`, `pre_t1_features.geojson`) under `demo/nepal_cascade/inputs/`, with an input audit recording every file hash. Of 4,414 mapped features, 2,092 carry damage dates at or before the cutoff; the rest are dateless, a fact that later decides the fate of flood memory.

Step 2 — context card. `agents/context_card.py` writes a redacted inventory (`card.json`, `card.txt`): what the inputs contain, with ISO date substrings redacted to `[date]` so reference metadata cannot smuggle time signals into prompts.

Step 3 — freeze. `scripts/freeze_study.py` copies (never edits) the inputs, prompts, and rules into the output directory and pins them in `BASELINE.sha256`. From here on, recomputation over moved inputs is refused: the hash check fails first.

Step 4 — compute the ledger (no key). The deterministic builder walks the study inputs feature by feature. Snapshot classifications (RCC-8 both directions, rectangle pairs, tiles, bands) come from `engine/qstr_core.py`; the continuity and band logic from `engine/gap_auditor.py`; flood-zone sets (418 zones, 518 informative) from `engine/fhz.py`. The eight ledger facts compute to: F1, 2,092 pre-cutoff destroyed features; F2, upstream share 0.5148; F3, channel-adjacent share 0.7285; F4, tributary-cluster rule HIT (34 features near the target reach); F5, upstream density 1.06× downstream (ground areas, not feature areas — the first density ratio computed this way came out 313269× and was caught by eye before freezing); F6, static flood outline as context; F7, target-reach share 0.0163, inactive; F8, inputs-audit clean. Replay command:

```
python3 -m agents.autoinsight --model-round on \
  --out /tmp/replay.json --prompt-log /tmp/replay_prompts
diff /tmp/replay.json demo/nepal_cascade/output/brief.json
```

The `--model-round on` flag replays the committed model cassette offline. The diff must print `REPLAY-IDENTICAL`.

Table 1: Frozen rankings. Fire probabilities reproduce to four decimals under the west-half mask; flood probabilities were fixed before ground truth opened. Full mechanism titles in the study briefs.

Study	Mechanism	Probability
Fire	suppression_then_restart	0.8425
	continued_spread	0.1140
	fuel_burnout	0.0419
	sustained_suppression	0.0016
Flood	impoundment_then_burst	0.9426
	upstream_slope_failure_elsewhere	0.0469
	downstream_only_flooding	0.0082
	safe_drainage	0.0009
	dam_holds	0.0009
	no_further_event	0.0005

Table 2: The model disagrees, the ledger decides. Both disagreements are published in the briefs; neither moved a frozen number.

Study	Model's top story	Ledger's top story
Flood	safe drainage first	impoundment-then-burst 0.94
Fire	smoldering_restart	suppression_then_restart 0.84
Shared:	fire model weighted the same two lead facts highest (run 0.95, clumping 0.9).	

Step 5 — score (worked). Active weights: F1 1.0, F2 1.0, F3 1.0, F4 2.0, F5 1.0, F6 0.25, F8 0.0; F7 inactive. Scores (prior 1.0 each, +0.5 caution on the quiet story only): impoundment $1+1+1+2+1 = 6.0$; slope-failure-elsewhere $1+1+1 = 3.0$; downstream-only $1+1+0.25-1 = 1.25$; safe-drainage $1+0-2 = -1.0$; dam-holds $1+1-2-1 = -1.0$; no-further-event $1+0-2-1+0.5 = -1.5$. Softmax gives raw 0.9427/0.0469/0.0082/0.0009/0.0009/0.0005, summing to 1.0001; the largest-residual correction takes 0.0001 off the top entry, yielding the frozen 0.9426. Probabilities are support strengths, not calibrated frequencies.

Step 6 — gate and brief. The default-deny verification gate checks every model-written sentence against the ledger's annotated numbers (entity ids masked first) and the brief is written to `demo/nepal_cascade/output/brief.json`.

Step 7 — the model's three rounds (key needed once, then cassettes).

Model: `muse-spark-1.2-contributor`. It was shown only the ontology plus the anonymized ledger. Round 1 drafted story `M_fluvial`. It *failed* the null red-team (thin framing) — rejected, kept on record. Round 2 failed the citation audit with 30+ short-id errors — rejected, kept on record, fixed by one instruction line. Round 3 (standing draft, top `M6`, a safe-drainage-family story) passed the null test — accepted. Re-recording uses `setup_study` with `--no-card`. Live runs need `BLINDSPOT_LIVE=1`; replay stays offline from committed cassettes (G14).

Step 8 — knowledge probe. `probe_model.py` asks the model about the event from anonymized evidence: "no knowledge," self-stated cutoff January 2026, predating the flood.

Step 9 — ground-truth check (after freeze only). `demo/nepal_cascade/gt_check.py` opens the quarantined T1–T2 delta: 754 newly destroyed features, channel-adjacent pattern, frozen top story consistent — MATCH, true-outcome rank 1, no re-tuning afterward.

6.2 Kalimantan fire: curate to both holdouts

Step 1 — curate. `demo/fire_study/curate.py` bins the stored FIRMS scene into tenth-degree floor-binned cells: 6,142 detections, 5 days (September 16–20), 30 cells. The cutoff (September 20) is derived from the data's maximum date at curation time, never typed into prediction code. Later September days ride along in `timeline.json` as held-out ground truth, unreadable from the prediction path.

Step 2 — card, Step 3 — freeze. Same tools as flood (card, then `freeze_study.py` + `BASELINE.sha256`). Known limitation, stated in the log: the fire setup round ran `--no-card` because card-carrying prompts fail their own audit on real inputs.

Step 4 — compute the ledger (no key). The spec-driven builder (`compute_ledger_from_spec`) evaluates the six fire facts over role-built cell lists: FF1, 6,142 detections; FF2, 5 distinct burn days; FF3, 30 cells with fire; FF4, 6,126 detections clumped near hot cells (cluster op scoped by reference role and filter — one of two generic op extensions the fire study forced, the other being the seconds-to-days scale divisor in time statistics); FF5, longest inter-burn gap (continuity signal from the day-to-day auditor); FF6, study-box area as context. Replay command:

```
python -m agents.autoinsight --inputs demo/fire_study/inputs \
  --rules demo/fire_study/rules.json \
  --domain demo/fire_study/domain.json \
  --out /tmp/fire_replay.json --model-round off
diff /tmp/fire_replay.json demo/fire_study/output/brief.json
```

Step 5 — score (worked). Supports: restart story FF2 (1.0) + FF4 (2.0) + FF5 (1.0); spread story FF1 + FF3; burnout story FF4 minus FF2; stay-out story FF6 (0.25) minus FF2 and FF4, plus the 0.5 caution. Scores: 5.0, 3.0, 2.0, −1.25. Softmax gives raw 0.8424/0.1140/0.0419/0.0016, summing to 0.9999; the largest-residual correction adds 0.0001 to the top entry, yielding the frozen 0.8425. Note the mirror image of the flood correction: the machinery corrects up or down by the same rule, and both directions appear in these two studies.

Step 6 — gate and brief, as flood.

Step 7 — temporal check. The ledger saw nothing past September 20; the held-out late-September days confirm suppression then restart. The check passes when the Step-5 top mechanism matches that outcome — it does, without ever reading the held-out days.

Step 8 — spatial check (re-run, not prose). Mask the eastern half at the median detection longitude (3,071 of 6,142 kept), recompute with the same frozen rules over `/tmp/fire_holdout/inputs`: identical ranking, identical probabilities (scoring depends on active facts, not magnitudes), every gate still passing on half the map.

Step 9 — the model's round. Same model, `--no-card`: six mechanisms (`burnout_suppression`, `smoldering_restart`, `spread_downwind`, `containment_hold`, `flareup_quiet`, `cessation`), weights highest on the unbroken run (0.95) and hotspot clumping (0.9), the same two lead facts the frozen brief leans on. Top: `smoldering_restart` with null PASS on the quiet story, disagreeing with frozen on top. Replay:

```
python -m agents.setup_study --inputs demo/fire_study/inputs \
  --domain demo/fire_study/domain.json \
  --frozen-rules demo/fire_study/rules.json \
  --brief demo/fire_study/output/brief.json \
  --out-dir /tmp/fsetup --no-card
```

6.3 Live reading: pull to page, no model at all

Step 1 — pull. Operator keys in environment only (never files, cassettes, or logs). Five days of NOAA-20 VIIRS for the study box, October 26–30: 675 detections (25/12/12/476/150), bounding-

box order verified, source matching the curated September data. The CSV lands in `/tmp` first, and a copy ships beside the snapshot so every number stays checkable. The pull never enters `data/` or the frozen study.

Step 2 — score with frozen rules. Role-building, ledger spec, and mechanism scoring run exactly as in the fire Step 4–5 over the live slice; the ranking must equal the frozen ranking within 0.00005 — it does, identically. No language-model call exists anywhere in this procedure.

Step 3 — pattern match and recall. The live signature is looked up against the stored pattern table: 60 episodes, 32 labelled, and today's shape sits in the Iberia-led pattern at 4 labelled of the 5 the bar requires — so the counts tier stays silent and the page says so. The new recall tier then shows what silence hides: the four same-shape windows (September 24–27 over Iberia, all restarted, with their detection counts), plus the two it refuses to call known — same shape, but past their scenes' last observed day, so kept unlabelable rather than guessed. This is the two tiers disagreeing honestly: one counts to four and stops, the other shows its work.

Step 4 — snapshot and page. `make_demo2.py` with `--live-pull` on the pull CSV writes the snapshot; the page carries the timeline (study window, hidden test, live days dated to October 30), the no-forecast boundary, and the conditional November outlook. When rain arrived after the last pull, no step changed: the outlook had named rain as its falsifier, and the next pull verifies it the moment daily counts fall.

7 Tiered memory and the self-evolving agent

Two frozen studies cannot teach a system anything — a pattern "seen twice" climbing a promotion ladder would be noise with a ceremony around it. Memory therefore begins by manufacturing honest cases: a rolling-origin backtest slides the cutoff across the stored fire history, runs the frozen prediction on pre-cutoff data only, and labels the outcome with the preregistered rule `restart-noisefloor-v1` (a restart inside the window supports the restart story; a quiet window supports the stay-out stories). Across fourteen scene files from eleven regions this yields 60 episodes, 32 of them labelled — all restart, because no archived scene has yet gone quiet (the August window, fetched as a quiet candidate, burned).

Memory is four tiers. T0 is the run itself, today's ledger and brief, pinned by hash. T1 is episodic: one append-only row per run, holding the case fingerprint, the frozen ranking, and — recorded later, never revised — the outcome and any operator verdict with its written reason. T2 is semantic: a pattern table that is *not stored at all* but recomputed on demand as a pure function of T1, so it cannot be tampered with silently. Below T2 sits the tier an operator actually looks at, case recall: given a window's facts, the five nearest labelled past windows, ordered by how many signature facts differ, each shown with its scene, cutoff, outcome window, and what followed — every step away named, so a near-match says how it differs, not just that it does. T3 is procedural: the rules, which change only by versioned amendment a human approves. All three reading tiers take an `as_of` date and see only episodes whose outcome window ended before the case's own cutoff; deleting every row a case could not have known leaves its brief byte-identical (gate G20, the memory counterpart of the leakage gate). Windows with no usable label yet are never displayed as known — same-shape ones are counted beside the results, so the missing quiet case stays visible instead of buried.

A pattern's signature is a short hash over binned fact values only: gap dictionaries project to binned day-gaps, raw floats (areas, magnitudes) leave the key entirely, because an early key fingerprinted every slice and produced 48 patterns of one. After the approved retune, 48 patterns collapse to 22 with cross-scene grouping, growing to 26 distinct shapes once the three targeted pulls

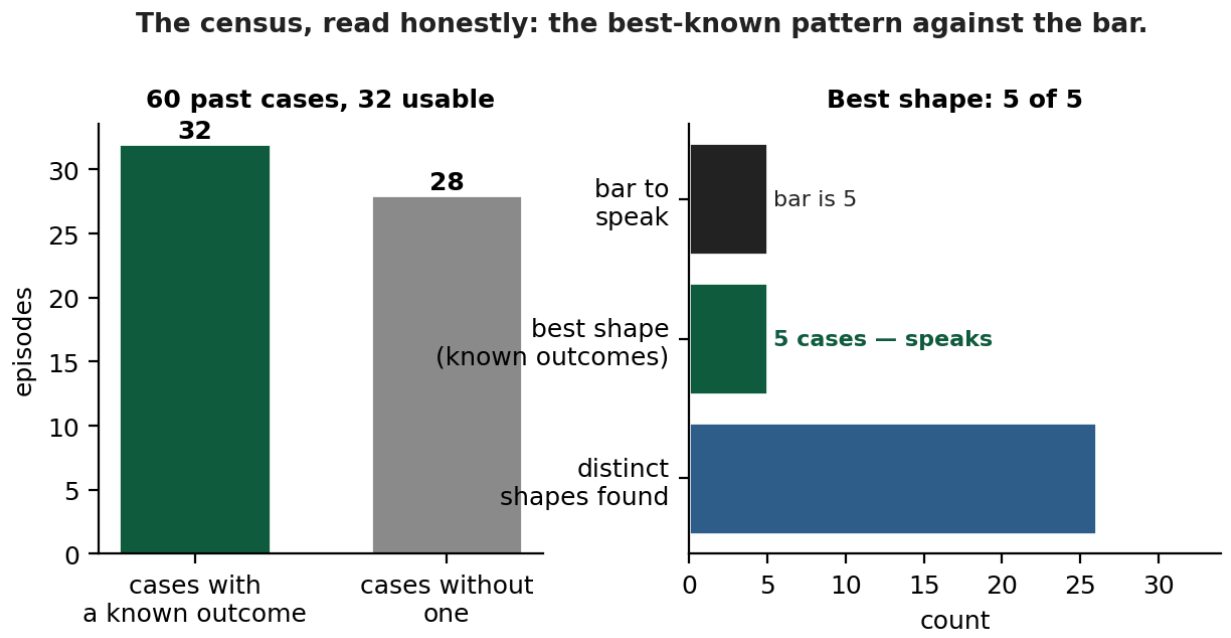


Figure 5: The census at the silent era's end, in one chart. Left: of 60 manufactured past cases, 32 have known outcomes (all restart). Right: they form 26 distinct shapes, and the best-known shape holds 5 of the 5 cases the preregistered bar requires — Amazon, Texas, the western US, and two Kalimantan windows, three continents in one shape. The chart that once read "one short" now reads met.

land — and for months the strongest stood at four labelled cases against a speaking threshold of five (Figure 5, drawn from that era). Memory stayed silent then, for a verified reason rather than a promised one: with the section absent, briefs are byte-identical to the frozen baselines (G19). Then three targeted pulls (Section 9) carried one shape over the bar, and the silence proved it was real by ending: new briefs gain the labelled second ranking, computed below, with the frozen ranking untouched beside it.

When the bar is met, the adjustment is deliberately weak. For each mechanism, with Laplace smoothing $\alpha = 1.0$ over n labelled cases:

$$r_m = \frac{\text{hits}_m + \alpha}{n + 2\alpha}, \qquad s_m = \text{clip}(r_m - \tfrac{1}{2}, \pm 0.10), \qquad p'_m \propto \max(0,\, p_m + s_m), \tag{1}$$

renormalized, with the frozen ranking never mutated and the shift block printed into the ledger so the gate can check the prose against it. The null test (Figure 3, right) proves this math cannot manufacture skill from noise: shuffled labels score -0.130 against frozen, while a perfect-signal control gains $+0.270$. An honest side finding: once memory speaks, long noisy histories actively hurt — recorded as a candidate rule fix, not a quiet retune.

The evolver — the self-improving agent — may propose, test, and wait. A Proposer role with read-only tools and a hard call cap drafts single-change proposals with cited episode and trace ids; structural checks (real ids, one change, recurrence of at least three independent episodes) gate it through a bounded revision loop whose feedback carries only pre-cutoff text; a Skeptic replays the change against held-out episodes the Proposer never saw, on a three-way time split whose final fifth is never touched during selection; and a human approves or rejects with a written reason, with

Table 3: The backtest census behind Figure 5. Rule `restart-noisefloor-v1`; two empty scene files and eleven thin-history cutoffs skipped with printed reasons.

Item	Count
Scene files (eleven regions)	14
Episodes	60
Labelled (all restart)	32
Unlabelable	28
Patterns under v4 key	26
Best pattern (labelled)	5 of bar 5: speaks
Quiet scenes (mapping unexercised)	0

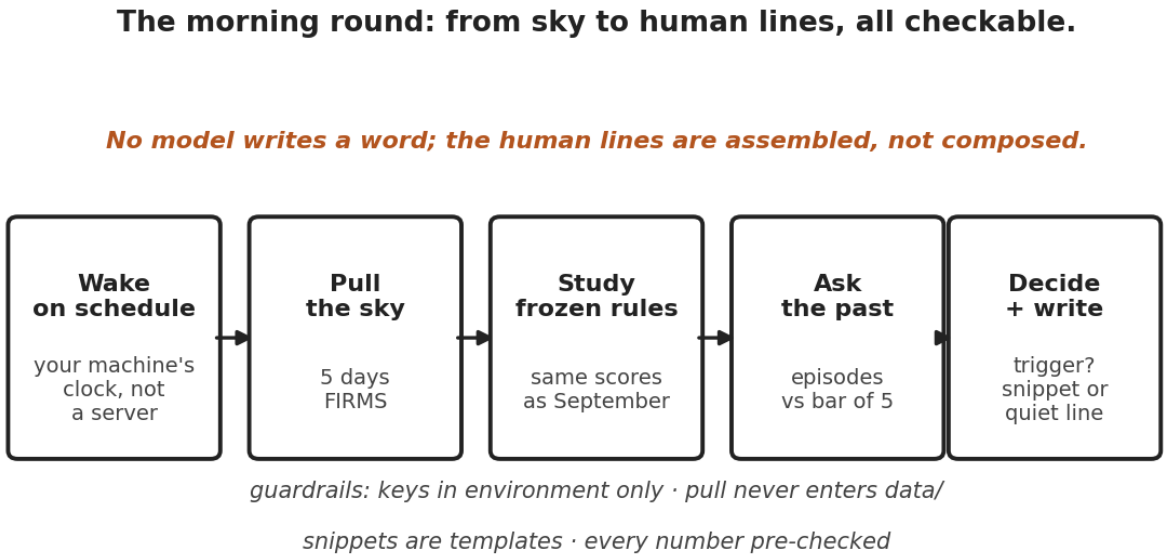


Figure 6: Five steps from sky to human lines. The guardrails are not aspirations: keys live in the environment, the pull never enters the frozen data, and no model writes a word of the digest.

approval minting a new version directory under a new tag. Instructions for the evolver's own agents evolve on a separate, cheaper track that cannot alter any scientific result. On today's data memory speaks on one shape while the evolver path still files nothing — zero candidates, no replay to clear, and the review tool proves it by refusing — which is the behaviour the gates pin (G22–G24).

8 The morning agent: a daily round, watched

Picture the operator's morning. Before coffee, before briefings, someone should have looked at last night's sky over the peat blocks and answered three questions: is anything new burning, does the past recognise it, and is today worth a full study? For the length of this project a person did that by hand. The morning agent is that round, automated — and automated in a way that cannot quietly become something else.

It lives in its own directory, `collab_agent/`, with one module per job: pulling the sky, studying

it, deciding, writing the human lines, and keeping the append-only digest. It wakes on the operator's own machine, on a schedule the operator owns — not as a hosted service, a boundary the project keeps deliberately (Section 12 says what would have to change before that line moves). Its keys travel in the environment only. If no key is present it logs "skipped" and exits cleanly: silence is reported, never faked, and a missing credential can never become an invented reading.

The round itself is five steps (Figure 6). First it pulls five days of satellite detections for the study box. Then it studies them with the frozen September rules — the same scoring code, the same weights, compared against the frozen ranking to within five hundred-thousandths; a difference would be reported, never adjusted. Then it asks the past: the live evidence shape is looked up against the 60 stored episodes, and the page says exactly where today stands against the bar of five — speaking, or silent with the count shown. Then it decides, and this is the step with the most design in it: four preregistered trigger rules, in fixed priority order, each with its reason written down before the runner existed. A day with twice the trailing median earns a study. A shape the archive has never seen earns one study. The fifth labelled case under one pattern is the morning memory speaks. A near-zero day after a matched pattern is logged as a possible first quiet — and nothing is labelled that morning, because a window is labelled only after it closes. If none fires, the agent writes one quiet line and stops; quiet days are evidence too, and the archive is starving for them. Finally it writes its two or three human lines — not composed by a model but assembled from a fixed template over ledger values, with every number checked against the ledger block before anything is written. A snippet that fails its number check is never published; the run fails instead.

Walk one real morning to see what this feels like. October 30: 150 detections against a trailing median of 18.5 — roughly eight times the normal level, past the two-times bar, so the spike rule fires with its reason attached. The frozen rules read suppression-then-restart at 84.25, identical to September. The past is asked: sixty episodes, 32 labelled, and today's shape sits at 4 labelled of 5 in its pattern — silent by the counts, stated as such, with the four matching Iberia windows listed beside the verdict and the two unjudgeable same-shape windows named as refused. The human lines go out naming the watch instruction for the week: daily counts near this level mean the burn run is unbroken. Every sentence of that morning traces to a number in the ledger, and the whole round replays byte-for-byte from the committed pull.

Two honest behaviours are worth naming because most automation hides them. The first is what happened the first time the runner met eight scattered synthetic detections in testing: they formed a shape the archive had never seen, so the agent opened a study instead of logging quiet — exactly what the preregistered priority order demands, and the test was corrected to expect it rather than the rule bent to expect quiet. The second is the missing-key morning: no credential, no drama, one logged line, exit zero. An agent that reports its own inability is one an operator can trust with the mornings it can handle.

9 Do fires rhyme across continents?

A pattern that repeats in one valley is local lore. A pattern that repeats in Iberia and Texas — different continents, different fuels, same hidden shape — is evidence the frozen rules carve nature at a joint. That is the cross-region study: eleven firegrounds on five continents (Kalimantan peat, Amazon rainforest, Cerrado savanna, Iberian scrub, Texas brush, Chaco, Mozambique, Zambia, northern Australia, the western US, Attica), each slid day by day through the same frozen rules — predict from everything before a cutoff, label from everything after — under one frozen outcome rule and one frozen pattern key. Sixty past episodes, 32 with known outcomes, 26 distinct shapes.

The count, recomputed from the memory store for this draft: 10 of the 26 shapes span more

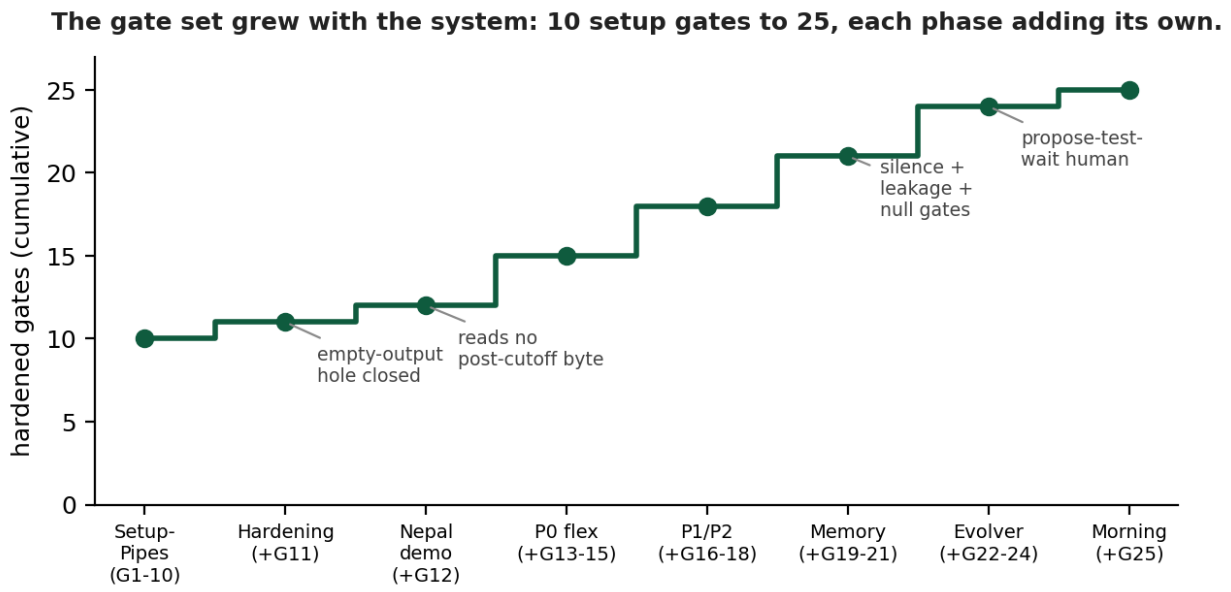


Figure 7: Ten setup gates grew to twenty-five. Each phase added the gates for its own new failure modes; the annotations mark what the additions caught.

than one scene. The leader holds five labelled cases of the five the bar requires — one Amazon, one Texan, one western-US, and two Kalimantan episodes, three continents in one shape — so memory speaks, and the silence section of this paper records when it ended. The old leader (Iberia plus Texas at four) still stands beneath it, unmoved; the new Iberian episodes formed different shapes, which is the bar working rather than bending. An operator reading this should feel both directions at once: the groups could be the beginning of general fire knowledge, or the key could be strict enough that such meetings stay rare.

So the study was designed to end, not to linger, and it ended like this. The diagnosis said the highest-value data in the world is narrow: extend the Iberian and Texan windows, and fetch one quiet window for a matched box. Three pulls landed; the extensions joined the Amazon trio's shape instead of the Iberian leader's — the data chose which pattern to complete, not the plan. The August window burned, so the quiet arm stays missing; that absence is reported, not hidden. The backfill path — curate into a pinned scene, rebuild the census cleanly, record the before/after — is the same agent code as the morning round, so the daily system and the learning system stay one system. Whatever the pulls say goes in the record as-is.

10 Hardened verification, and how useful it was

The short version: the gates earned their keep, twice over — once while the code was being written, once while the system runs on real data — and Table 5 lists the receipts. The rest of this section explains what "hardened" means, why the project needs it, and where its limits lie.

What hardened verification is, and why it exists. Most of this system was written with machine assistance, and machine-written code has a specific failure mode: it fails *silently and fluently.* It produces results that look structurally perfect — valid tables, valid briefs, green test suites — while being wrong underneath. A human reviewer reading that output nods along, because

Table 4: The hardened gate inventory. G3b is a cross-study variant of G3; G15 covers the P0 study-gate criteria; G16–G18 cover rebuild, restore, and ingest; G19–G24 cover memory and evolution; G25 pins the morning round.

Gates	What they enforce
G1, G8	Manifest discipline; input layouts conform
G2	Config complete (null means undecided, never defaulted)
G3, G3b	No hardcoded scenario answers in builder code
G4, G5	Null config exits non-zero; both pipelines exit 0 on real data
G6, G7	Output contract, inverse relations, shift invariance
G9	Frozen config under a git tag before any pipeline work
G10	Durable suite runs inside the gate (blocking)
G11	Content-vs-input: counts reconcile, scenario file reselected by rule
G12	Prediction reads no post-cutoff byte; prompts quarantined and re-audited
G13–G15	Cross-study genericity; frozen replay identical
G16	One-command rebuild: manifest, gates, seed, API, judge
G17, G18	Restore drill; raw-ingest watch
G19, G20	Memory silence and memory leakage (as-of filtering)
G21	Null red-team: shuffled labels must not beat frozen
G22, G23, G24	Evolution discipline; loop bounds; feedback leakage
G25	Morning digest deterministic; data/ untouched; frozen briefs unmoved

everything parses. "Hardened" means the checks do not read the output the way a reviewer does; they recompute it from an independent direction and compare. Hide all raw inputs and re-run: is the brief byte-identical? Re-derive the scenario file from the selection rule: is it the file the output claims? Shuffle every label: does memory still claim skill? A gate fails closed — a skipped check counts as failure — prints only claim identifiers, and the phase is not done until the full run reports ALL PASS. An independent judge then replays the durable suite from scratch (147 checks green at last observation).

The two moments: coding-time and execution-time. The twenty-five gates in Table 4 split into two moments, and they catch different defects. *Coding-time* checks run while the system is being built: property tests asserting math facts the code must obey (converse-symmetry on every relation pair, inverse snapshots everywhere), literal scanners forbidding scenario answers in builder code, contract tests on outputs. *Execution-time* gates run the built system on real data and demand proof: the leakage re-run with hidden inputs (G12), the frozen replay demanding identical bytes (G14), the content-vs-input reconciliation (G11), the memory silence/leakage/null trio (G19–G21), the restore drill (G17). Figure 7 shows the set growing from ten setup gates to twenty-five as each phase added the gates for its own new failure modes.

What the gates caught. Table 5 is the usefulness ledger: each row is a real defect, the moment that caught it, and what it was worth. Three rows deserve a sentence each. The composition-table rows were zipped against the wrong sort order, silently shifting every row after P0; every downstream inference looked structurally fine while being wrong, and only the symmetry plus worked-example checks noticed — the canonical case for why review-style reading is not verification. The vacuous-truth hole was worse: a structurally valid but content-empty result passed everything, so the fix (G11) was proven on a doctored but self-consistent control built from the wrong input file — the exact failure the original empty-output incident deserved. And the leakage gate caught nothing because there was nothing to catch, which is itself evidence: hiding all raw inputs and re-running produces byte-identical briefs, so the predictions provably read no post-cutoff byte. The

Table 5: What the verification caught, and when. "Coding" rows broke the build; "execution" rows failed a run on real data. Worth is the author's honest grade.

Defect	Caught	Worth
Composition-table rows shifted by sort order	coding: symmetry tests	very high
Empty result passing all structural checks	coding: G11 + doctored control	high
Model's short ledger ids (30+ cite errors)	coding: round rejection, 1-line fix	high
Post-cutoff leak search (absent, proven)	execution: G12 hide-raw re-run	high
Snapshot script printing "wrote" without writing	execution: read-back step	medium
Store timestamps breaking byte-stability	execution: restore drill	medium
Float-equality test blaming correct code	coding: fixed the test, not the code	medium
Prose tripping the literal scanner	coding: reworded, gate kept	low, real

null gate's verdict — memory loses to noise by 0.130 — is a result, not a hurdle: it is the number that lets this paper claim the adjustment cannot invent skill.

What plain process caught. Not everything belongs to the gates, and the per-phase log keeps the tags honest. The standing rules that emerged are boring and load-bearing: execute, don't eyeball (both one-character defects in this project were found by running, never by reading); read back every write; never trust a check built from the assumption under test; never weaken correct code to make a self-authored test pass.

What nothing here can see. Three limits are stated openly because they bound every claim above. First, the gates check the rule, not the world: if the data directory itself held the wrong files, every gate would still agree with itself; that final mile is human judgment plus the manifest. Second, gates assert structure and bytes, not framing choices: no gate checked the map viewport that once fitted a 172.8 km band instead of the feature, and prose is load-bearing in this repository down to single words. Third, cost is real: the full suite runs just over a minute, prose scans need tending with every new page, and each selection step against one validation slice spends a little of that slice's honesty — which is why the final fifth of history is never touched during selection and is read once, at the end, for the reported result.

The net assessment, after two studies and a memory arc: the gates were very high value for the kernel and pipelines (where silent wrongness is cheap to produce and expensive to notice), high value for memory and evolution (where the failure mode is fooling oneself), and a useful tax everywhere else. A flat result would still have been reported — Section 7 shows the machinery saying "nothing qualifies" four separate times — and that is the behaviour a reader should trust most.

10.1 Postscript: the falsifier fires during drafting

On October 30, with the November outlook on the page in conditional form ("if the dry regime returns"), news reports arrived that rain had eased the Kalimantan fire. The rain fell after the last satellite pull, so it sits outside every frozen number in this draft — which is precisely the arrangement the outlook specified: sustained rain moves the live reading out of the restart signature, and the next pull verifies it the moment daily counts fall. If confirmed, the quench becomes the archive's first quiet: the missing arm exercised at last, and the first test of the quiet mapping the labelling rule has never yet used. It is not the fifth label that lets memory speak — the Amazon-led shape already stands at five of five and speaks; the live reading belongs to the Iberia-led shape at four, and a confirmed quench under that shape would carry it over the bar with a quiet rather than

a restart, the hardest possible way to earn the fifth chair. A draft that predicts its own falsifier and then watches it arrive is either lucky or honest; the pull CSV will decide which, and whatever it says gets recorded as-is.

11 Limitations

A first draft is allowed cons, and this system has specific ones. The history is one-sided: 32 restarts, zero quiets, so every learned rate is a rate of restarting and the quiet mapping is untested code in the scientific sense. Neighbouring backtest cutoffs overlap heavily, so episodes are not independent no matter how the independence gap is set; the held-out splits are strictly later in time, which mitigates but does not cure this. The outcome rule is a proxy for truth, and operator verdicts are kept separate from it so the proxy can later be audited rather than trusted. Flood memory is empty for a data reason (two dated damage days against 8,368 dateless features) that no method can argue with. The flood spatial holdout is pending. Setup with the context card still fails its own prompt audit on real inputs; both committed rounds ran without the card. One recorded deviation bends the freeze discipline: the fire rules were amended after first freeze with weights locked and outcomes accepted, counted as amendment rather than tuning, with per-amendment rationale on file — future studies must freeze the learnings pass explicitly instead. Concretely, the amendment moved weight placement, rebound fact FF4 to the scoped cluster operation, and tightened the prompt contract; no weight value changed, no outcome was re-decided, and the brief bytes before and after are identical except the intended additions. A reviewer suspecting the outcome was gamed should check exactly that diff: the verdict 0.8425 does not move across it. And there is no forecast anywhere in the system: every forward-looking sentence is a validated pattern pointed at today, extended one careful step.

12 Future work: demonstrations other systems will lack

Each item below ends in a demonstration that is hard to copy precisely because it requires the machinery above to already exist.

The day memory spoke — and the day a second shape speaks. Memory already spoke once: the Amazon-led shape crossed to five of five, and briefs gained the labelled second ranking a null test had proven cannot invent skill, with the silence proof and the speech proof side by side. The demonstration other systems will lack is the second crossing: the Iberia-led shape stands at four, and whichever closed window carries it over — a restart or, harder, the first quiet — shows the transition a second time, on demand rather than in hindsight. No system without a preregistered bar and a byte-identity silence proof can show either crossing honestly.

Flood dated observations and the missing spatial cell. Collecting dated damage observations plus a preregistered flood outcome rule unlocks both flood memory and the masked half-valley re-run. The demo is the fourth cell of Figure 4 flipping from PENDING to MATCH (or mismatch, reported as-is) — a before/after no re-tuning regime can stage.

An evidence-strength gate. Once memory speaks, long noisy histories actively hurt (null -0.130); a gate that weighs evidence by recency and independence, preregistered before it is measured, becomes a demo of learning that refuses to overfit — the rarest kind.

The quiet mapping. The first archived quiet — possibly the October rain quench — exercises the labelling rule's other arm, tests the stay-out stories, and gives the Skeptic its first real replay. The demo is memory learning that fires can end, on record.

Card-carrying setup rounds. Fixing the audit alignment for card text unblocks setup rounds that read the full redacted inventory. Fresh cassettes, and a new prompt-hash lineage to show for the fix.

A third scenario, then the full daemon. A third domain (storm surge is the natural candidate: same gap shape, different physics) proves the catalog first. Only then does the fuller event-driven loop earn its place: a scheduled trigger that slides cutoffs over arriving data and starts a hill-climbing step, with every proposal still waiting for a human. The morning agent of Section 8 is already that loop's first, deliberately small form — running on the operator's machine, on the operator's schedule, and only reading. The demo is an operator's morning brief that wrote its own audit trail overnight. Unfreezing anything more remains an evidence decision, not an anticipation.

13 The six stories: what the showcase pages show

The results above live twice: once in this draft, once as six story-first pages off the demos index, written for a visitor who will never read a gate table. Each pairs what the engine saw inside a sensing gap with the proof it could not have cheated, quoting the command and its output verbatim. The flood trio: the blindfold reveal (the 0.9426 verdict spoken without the aftermath, then 754 sealed features matching it), the triage map with the overruled model (195 assets sorted into the maybe-band; M_fluvial rejected, forty-two citation errors, M6 passed yet outvoted 0.9426 to 0.0009), and the teleport-proof algebra (the symmetry and vacuity gates that caught real bugs before the freeze). The fire trio: the zombie fire read from absence at 0.8425 with the future sealed, the halved map returning identical probabilities to six decimal places of zero drift, and the awakening — five windows across three continents ending the silence, then the model-less morning watch naming its rain falsifier in advance. The pages add no new claims; every number on them already appears in the sections above, which is precisely why they can be told like stories.

14 Conclusion

BlindSpot's claim is small enough to check: qualitative relations over sensing gaps, scored by frozen weights, framed but never moved by a language model, verified by gates that have caught real defects and state their blind spots in writing. The fire study passes two holdouts; the flood study passes one with the other openly pending; memory spoke once its bar was met, after staying silent for verified reasons while it was short; the evolver proposes, tests, and waits. The rain that arrived during drafting gets the last word it earned: the system named its falsifier in advance, and the next satellite pull will say whether the pattern held. Whatever it says goes in the record as-is — that is the whole design, in one sentence.

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